Reviewer Pack — Minimal Rank of Geometric Phases of Manifolds Bounds Communi...

Entry 711a063377d7 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

2026-05-28 08:06:06 UTC

Minimal Rank of Geometric Phases of Manifolds Bounds Communication Complexity for XOR

Entry ID: `711a063377d7` **Verdict**: `INCONCLUSIVE` **Recorded**: 2026-05-28 08:06:06 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Topology (Geometric Phase Theory) **Field B** (complexity object): Communication Complexity

Statement:

> ['For a given n-bit Boolean function f, define the geometric phase of its support manifold M as the minimal rank of an orthonormal basis for the tangent space at each point of M. Then, for any t > 0 and n ≤ 40, the communication complexity CC(XOR) for XORing two n-bit inputs is upper bounded by Θ(2^(n/4 * min_r(r! * log(1/r), log(t))).', 'Equivalently, if there exists an n-bit function f such that its geometric phase rank is less than 2^(n/4 * min_r(r! * log(1/r), log(t))), then CC(XOR) > Θ(log(t)).']

Rationale (proposer's reasoning):

> ['Geometric phases capture the topological properties of manifolds and may provide insights into communication complexity, which deals with the minimum number of bits required to transmit information between parties. A lower bound on geometric phase rank could imply a lower bound on communication complexity.', 'This conjecture leverages the mathematical tools from algebraic topology to study communication complexity, potentially revealing new connections between these two fields.']

Taxonomy category: `geometric_phase_communication_complexity` (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): `370d11e37c1ecd9c`

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

> The conjecture is supported if, across 30 random seeds, the geometric phase rank of all n-bit Boolean functions is less than or equal to 2^(n/4 * min_r(r! * log(1/r), log(t))) for all t > 0 and n ≤ 40. The conjecture is falsified if any seed produces a function with geometric phase rank greater than 2^(n/4 * min_r(r! * log(1/r), log(t))).

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM₁	LLM₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.70	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "geometric phase theory" AND "communication complexity" AND "minimal rank" - "tangent space" IN algebraic topology AND communication complexity AND XOR function" - "XOR" communication complexity "geometric phase" rank bound

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)
Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution**: rc=124, elapsed=240.5s

5.1 Generated Python source

```python
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def factorial(n):
        if n == 0 or n == 1:
            return 1
        result = 1
        for i in range(2, n + 1):
            result *= i
        return result

    def geometric_phase_rank(n):
        # Placeholder function to compute the geometric phase rank
        # This is a dummy implementation and should be replaced with actual logic
        return random.randint(1, 2**(n//4))

    def communication_complexity(rank):
        return 2**(n/4 * min(factorial(i) * math.log(1/i), math.log(t))) if rank > 0 else float('i

    n = random.choice([5, 10, 15, 20, 30, 40])
    t = random.uniform(1, 100)
    rank = geometric_phase_rank(n)
    cc = communication_complexity(rank)

    return {
        "metric_name": "Communication Complexity",
        "metric_value": cc,
```

```
        "instances_tested": 1,
        "conjecture_holds": rank <= 2**(n/4 * min(factorial(i) * math.log(1/i), math.log(t))) for
        "counterexample": "" if rank <= 2**(n/4 * min(factorial(i) * math.log(1/i), math.log(t)))
    }

if __name__ == "__main__":
    seeds = [int(arg) for arg in sys.argv[1:]] or [random.randint(1, 1000) for _ in range(30)]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {{\"seed\": {seed}, \"metric_name\": \"{result['metric_name']}\", \"metric_
        results.append(result)

    mean = sum(r['metric_value'] for r in results) / len(results)
    std_dev = math.sqrt(sum((r['metric_value'] - mean)**2 for r in results) / len(results))
    support_fraction = sum(1 for r in results if r['conjecture_holds']) / len(results)

    if all(r['conjecture_holds'] for r in results):
        print(f"RESULT: SUPPORTED mean={mean} std={std_dev} support_fraction={support_fraction}")
    elif any(not r['conjecture_holds'] for r in results):
        first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result['conj
        print(f"RESULT: FALSIFIED counterexample=\"Rank exceeds bound\" first_failing_seed={first_
    else:
        print("RESULT: INCONCLUSIVE mapping_undefined")
```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
TIMEOUT after 240s
```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

> skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

> The test timed out before producing data, which means we cannot confirm whether the conjecture is supported or falsified. | next: Run the test again with increased time limits to ensure it completes and produces results.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	11977
2	preregistration	ollama_remote	glm4:latest	0	0	6630
3	novelty	ollama_remote	glm4:latest	0	0	4648
4	novelty	ollama_remote	glm4:latest	0	0	10189
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14240
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12557
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12041
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9096
9	judge	ollama_remote	glm4:latest	0	0	13766

Totals: 0 input tokens, 0 output tokens, ~$0.0000 cost, 95143 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/711a063377d7.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/711a063377d7.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)
Replay tarball: `research/replay/711a063377d7.tar.gz` (if generated)